

研究报告：基于 Reti-Pioneer 的单调质量路由与端点感知多任务学习扩展

日期： 2026-09-15 (review-fixes)

SYNTHETIC metrics ≠ clinical AUROC

封面信息

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1. 摘要

Reti-Pioneer 发表文本描述多任务筛查，但发布训练代码为疾病独立二分类循环，质量权重固定 1/0.5/0。本仓库实现单调有界质量路由、掩码多任务 BCE、endpoint-aware cross-cohort evaluation。校准/ECE/按病种 DCA 为评估框架（非新颖性主卖点）；NB@0.10 仅示意。当前数值为 SYNTHETIC 流水线自检，不得作为临床性能。

2. 创新边界

| ID | 创新 | 状态 |

3. 方法映射

QualityAware.quality_router · ensemble=released_code · label_map.Endpoint ·
split.calibration_idx · masked BCE · evaluate --paper-mode ·
configs/ablation_e0..e4.yaml。详见 docs/paper/manuscript.md。

4. 结果（SYNTHETIC）

Arm · Eval · AUROC · AUROC_D · ECE · Cal ECE · NB@0.10 · n · Note

<figure><img

src=" i HZlcnNpb24zLjEwLjksIGh0dHBz0i8vbWF0cGxvdGxpYi5vcmcvJkbTWQAAAAIwSF I zAAAuIwAALiMBeKU/dgABAABJREFUeJzs3QWYVGUbxvGH7u5uCWnpBgVFEBQQAQEBAeI SQBBFRDBQEKUUEASku0ukUwmRkJDu7ma+63n9Zp i ZnZmd3Z3ILLv/33UNTJw5c+b0znuf541ms9IsAgAAAAAAAAAAAAAFgkuIUfDAAAAAAAAAAAAAACgCLAAAAAAAAAAAAAALAUARYAAAAAAAAAAAAAABYigALAAAAAAAAAAAAALEWABQAAAAAAAAAAAAAJYiwAIAAAAAAAAAAAAAABLEWABAAAAAAAAAAAAACA pQi wAAAAAAAAAAAAAAwF I EWAAAAAAAAAAAAAAGApAi wAAAAAAAAAAAAAAACwFAEWAAAAAAAAAAAAAAWIoACwAAAAAAAAAAAAACxfgAUAAAAAAAAAAAAACWIsACAAAAAAAAAAAAAASxfgAQAAQMBt3LhR8uTJl6lSpZJR0YxhwELtsG3335b4sSJl+PHj4+083/L I i3SpEkT i Rs3bqT+ngjqzJkzUqVKFUmSJl m0atVKHj16xGwKg0vXr8v06d0j3Ly02WyybNkyqVWrlsSIEUNWr14d0PGzviKq2L17t7Rt21YSJkwon376qdWTg1DQv91Sp04tuXPnNueTT8PSpUv l x l kTT+WzAAAAEPEPYAEAAEDAG4Heeust+eeff+TixYvSvn17cx/PjhUrVkjx4sWIWbNmV8KQuD27dsyZswYKVKkiJQtW1YmT54s9+/fj3Tz807duyasUqxYMSIVqpRMmjRJ7t27Z/Vk4Snr3r27/P777yZwoev9uHHjWAZh8PjxY/n++++lUKFCcuPGjSgzL69cuSJDhgyR5557Tl555RVZsGCBmReBxvqKy0zBgwcybdo0qVChghQsWNAEIG7dumX1ZCEU9G82/dvtwoULcuDAAfM3nf5t9zQULVpUPvnkE87pAAAAQIAFAAAgMvvr78kf/78kjZtWokWLVpAb96uqtTQysmTJx2PtSFIr8bEs1HNQisaVKtWTf744w+JCFatWmXWX70i3tu62KFDh4B/7s2bNyV58uQePy9mzJiSPn16c6V+RDFo0CDJkCGDqUSxc+d0iaxGjBghGTNmI0bNm8uff/5p9eTAQjt27HB5HJnX+/CmdZW63x89erSsXLIS3n33XYkKunXrZvab+v+hQ4fC9bNYXxFTZw4UTJnz i wNGjSQdevWWT05CMDfjs4hPv2b7tKlS+E+XzVAq0vPrFmzTBD777//DvfPBAAAQMRFBRYAAIBITK+C1B8Az549K9u3bzdVNdyVLl3adDXi6dawYUPTSK9XxMWKFcuVz0yZMqVky pTJ8ViDBzodiLj27t0rb7zxhqImoRUNIhIN10j6qxUBtHE1QYIEQYb5+eefTQNsIP3000/mynx32tCpP+SfPn1a5s+fh9DPDluXX35Z9uzZl//++69pSlqsy pcvL7t27ZLDhw+bIAu i Lj0uOdMGL4QuWKHzUi vZbNiWQXLkyBFIZq0e4xw/ftzMg0SJEoXrZ7G+IrlqWbKk6WZGu8nSymh4tmkVrujRnzQX6N90+rfd06Ddz65du9b87ajrVUQ6zwYAAMDTFfMpfx4AAAAsoo17U6ZMkZw5c7o837p1a7+6ir l 8+bL0799fvvvu05/DaYWKqV0nSosWLUxDv75HS/Mj4tGGu759+5qrZx89eiQRWFz48aVly5amm5z0nTu7vHbnzh2zXg4YMCBgpfc1SwlPDZDffvutRETOlBE333wzwk5nWBuUMBxv169esHu jxB5

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MWzYM
8axxr
aqSaF
TKJFi
yy+/L
0qITp
DWJAR
FXROn
y1qPE

5. 讨论与局限

须区分 published multitask vs released independent loops; diabetes_related
不得进临床主表。待补充：真实 ODIR/BRSET 像素、GPU 特征。工程侧已补：val→test
阈值冻结、patient-level bootstrap CI、E5 门控、 λ_q 、intervention smoke。

6. 验收

见 docs/chatgpt-runs/2026-09-13-review-fixes/ACCEPTANCE.md。



fig2_ablation_bars

1.0
0.8
0.6
0.4
0.2
0.0

AUROC

